

ME 639 – Introduction to Robotics

Homework 1: Foundations & Rotations

AY 2026–27 | Madhu Vadali | IIT Gandhinagar

Due: [Thursday, September 3rd, 2026]

AI Use Policy Reminder

AI tools are permitted as *learning aids*, not answer generators. Any submitted work that involved AI must include a completed **AI Use Note** (see end of this document). Undisclosed use is an academic integrity violation. Before submitting anything AI touched, close the tool and explain the concept in your own words — the *Explain-It-Without-the-Response* test. Where a specific part below is marked **No AI**, that part must be done without AI assistance of any kind. Full policy: *AI Class Policy.pdf*.

How this assignment is graded. Part 1 (pen and paper) is graded on *effort*, not on arriving at the correct final answer. Struggling with a problem is fine, and expected, as long as you show your reasoning and it is clear you engaged with the ideas. Part 2 (implementation) is graded on what you built and what you can explain about it, not on producing a polished/finished app.

Part 1 – Pen and Paper

Show your work – a correct final answer with no reasoning shown earns little credit; a genuine, well-reasoned attempt that doesn't fully close out earns most of the credit.

No AI on this part

Work this Part out by hand, without any AI assistance; Grading on this part is for effort and a genuine attempt, not for arriving at the correct final expression. You may however check your answer after putting in genuine effort to solve it. If AI is used to check, please *report and reflect*.

Problem 1. Vector and cross-product identities

Source: Murray, Li & Sastry (1994), Ch. 2, Exercise 1

Let $a, b, c \in \mathbb{R}^3$ be 3-vectors and let $\cdot$ and $\times$ denote the dot product and cross product in $\mathbb{R}^3$. Verify the following identities:

(a) $a \cdot (b \times c) = (a \times b) \cdot c$

(b) $a \times (b \times c) = (a \cdot c)b - (a \cdot b)c$

Problem 2. Frames, rotation matrices, and what they represent

Source: Lynch & Park (2017), Ch. 3, Exercise 3.1(a–h)

In terms of the $\hat{x}_s, \hat{y}_s, \hat{z}_s$ coordinates of a fixed space frame $\{s\}$, the frame $\{a\}$ has its $\hat{x}_a$ -axis pointing in the direction $(0, 0, 1)$ and its $\hat{y}_a$ -axis pointing in the direction $(-1, 0, 0)$, and the frame $\{b\}$ has its $\hat{x}_b$ -axis pointing in the direction $(1, 0, 0)$ and its $\hat{y}_b$ -axis pointing in the direction $(0, 0, -1)$.

- (a) Draw by hand the three frames, at different locations so they are easy to see.
- (b) Write down the rotation matrices R_{sa} and R_{sb} .
- (c) Given R_{sb} , how do you calculate R_{sb}^{-1} *without* using a matrix inverse? Write down R_{sb}^{-1} and verify its correctness using your drawing.
- (d) Given R_{sa} and R_{sb} , how do you calculate R_{ab} (again without using matrix inverses)? Compute the answer and verify it using your drawing.
- (e) Let $R = R_{sb}$ be considered as a transformation operator consisting of a rotation about $\hat{x}$ by -90° . Calculate $R_1 = R_{sa}R$, and think of R_{sa} as a representation of an orientation, R as a rotation applied to R_{sa} , and R_1 as the new orientation after the rotation has been performed. Does the new orientation R_1 correspond to a rotation of R_{sa} by -90° about the world-fixed $\hat{x}_s$ -axis or about the body-fixed $\hat{x}_a$ -axis? Now calculate $R_2 = RR_{sa}$. Does R_2 correspond to a rotation of R_{sa} by -90° about the world-fixed $\hat{x}_s$ -axis or about the body-fixed $\hat{x}_a$ -axis?
- (f) Use R_{sb} to change the representation of the point $p_b = (1, 2, 3)$ (given in $\{b\}$ coordinates) to $\{s\}$ coordinates.
- (g) Choose a point p represented by $p_s = (1, 2, 3)$ in $\{s\}$ coordinates. Calculate $p' = R_{sb}p_s$ and $p'' = R_{sb}^T p_s$. For each operation, should the result be interpreted as *changing coordinates* (from the $\{s\}$ frame to $\{b\}$) without moving the point p , or as *moving the location* of the point without changing the reference frame of the representation?
- (h) An angular velocity ω is represented in $\{s\}$ as $\omega_s = (3, 2, 1)$. What is its representation ω_a in $\{a\}$?

Problem 3. Composing three basic rotations

Source: Lynch & Park (2017), Ch. 3, Exercise 3.2

Let p be a point whose coordinates are $p = \left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{2}}\right)$ with respect to the fixed frame $\hat{x}$ – $\hat{y}$ – $\hat{z}$. Suppose that p is rotated about the fixed-frame $\hat{x}$ -axis by 30° , then about the fixed-frame $\hat{y}$ -axis by 135° , and finally about the fixed-frame $\hat{z}$ -axis by -120° . Denote the coordinates of this newly rotated point by p' .

- (a) What are the coordinates p' ?
- (b) Find the rotation matrix R such that $p' = Rp$ for the p' you obtained in (a).

Problem 4. Proving $R_a^b R_b^c = R_a^c$

Source: Lynch & Park (2017), Ch. 3, Exercise 3.4

In this exercise you are asked to prove the property $R_a^b R_b^c = R_a^c$ used in class. Define the unit axes of frames $\{a\}$, $\{b\}$, and $\{c\}$ by the triplets of orthogonal unit vectors $\{\hat{x}_a, \hat{y}_a, \hat{z}_a\}$, $\{\hat{x}_b, \hat{y}_b, \hat{z}_b\}$, and $\{\hat{x}_c, \hat{y}_c, \hat{z}_c\}$, respectively. Suppose that the unit axes of frame $\{b\}$ can be expressed in terms of the unit axes of frame $\{a\}$ by,

$$\begin{aligned}\hat{x}_b &= r_{11}\hat{x}_a + r_{21}\hat{y}_a + r_{31}\hat{z}_a, \\ \hat{y}_b &= r_{12}\hat{x}_a + r_{22}\hat{y}_a + r_{32}\hat{z}_a, \\ \hat{z}_b &= r_{13}\hat{x}_a + r_{23}\hat{y}_a + r_{33}\hat{z}_a.\end{aligned}$$

Similarly, suppose that the unit axes of frame $\{c\}$ can be expressed in terms of the unit axes of frame $\{b\}$ by

$$\begin{aligned}\hat{x}_c &= s_{11}\hat{x}_b + s_{21}\hat{y}_b + s_{31}\hat{z}_b, \\ \hat{y}_c &= s_{12}\hat{x}_b + s_{22}\hat{y}_b + s_{32}\hat{z}_b, \\ \hat{z}_c &= s_{13}\hat{x}_b + s_{23}\hat{y}_b + s_{33}\hat{z}_b.\end{aligned}$$

From the above, prove that $R_a^b R_b^c = R_a^c$.

Problem 5. Properties of rotation matrices and $\mathfrak{so}(3)$

Source: Murray, Li & Sastry (1994), Ch. 2, Ex. 3(c) & Ex. 4(a-c)

Let $R \in SO(3)$.

- (a) **[Recall from lecture]** Show that for $v, w \in \mathbb{R}^3$,

$$R(v \times w) = (Rv) \times (Rw).$$

This was derived in class – write the argument again in your own words (no need to search for a new proof).

- (b) For $\omega \in \mathbb{R}^3$, show that

$$R \hat{\omega} R^\top = \widehat{(R\omega)}.$$

- (c) Let $R = \begin{bmatrix} r_1 & r_2 & r_3 \end{bmatrix}$ be a rotation matrix written in terms of its columns. Show that

$$\det R = r_1^\top (r_2 \times r_3).$$

- (d) Show that $\mathfrak{so}(3)$ (the set of 3×3 skew-symmetric matrices) is a vector space. Determine its dimension and give a basis.

Problem 6. The planar rotation group $SO(2)$

Source: Murray, Li & Sastry (1994), Ch. 2, Exercise 10(a,c)

Let $SO(2)$ be the set of all 2×2 orthogonal matrices with determinant equal to $+1$.

- (a) Show that $SO(2)$ can be identified with $\mathbb{S}^1$, the unit circle in $\mathbb{R}^2$.
 (b) Show that for $R \in SO(2)$ and $\hat{\omega} \in \mathfrak{so}(2)$,

$$R \hat{\omega} R^\top = \hat{\omega}.$$

Only fill in the note below if you used AI on a non-flagged part of Part 1 (most students will leave this one blank).

AI Use Note (required if AI was used on Part 1)

Tool used:

What I asked it:

How I used the output (and what I changed/verified myself):

Part 2 – Implementation & Visualization (MuJoCo/ROS)

AI usage is *Optional* on this part

AI use (vibe coding included) is *optional*. It is expected and encouraged to adhere to the standard disclosure policy above: document and report your prompts and what you changed/verified. Please *report and reflect*.

Starter code

`code-templates/hw01-mujoco-rotations/` in this repository. See that folder's `README.md` for setup instructions.

Problem 7. Task 1 – Rotation sandbox: does order matter?

Using the provided asymmetric-body MuJoCo model (`model/asymmetric_body.xml`), extend `scripts/01_rotation_sandbox.py` so that a user can apply a sequence of elemental rotations (about $\hat{x}$, $\hat{y}$, $\hat{z}$) to the body, *choosing at each step* whether the rotation is about the **current (body) frame** or the **fixed (space) frame**. Reproduce, visually, the non-commutativity you proved symbolically in Problem 3: show a sequence of two rotations that gives a visibly different final orientation depending on which frame each rotation is taken about. Capture a short screen recording (10–20s) of both cases side by side.

Problem 8. Task 2 – Verifying the skew-symmetric identities in simulation

Extend `scripts/02_verify_skew_properties.py` to spin the body under a time-varying angular velocity, log the rotation matrix $R(t)$ that MuJoCo simulates, and numerically check the two identities from Problem 5: $R(v \times w) = (Rv) \times (Rw)$ and $R\hat{\omega}R^T = \widehat{(R\omega)}$, for several random v, w, ω at several simulated time steps. Report the numerical residual (it should be near machine precision) and briefly explain, in your own words, *why* simulating this doesn't itself prove the identity – what would it mean if the residual were small but not exactly zero?

Problem 9. Task 3 – Current frame vs. fixed frame in ROS/rviz

Using the ROS 2 package skeleton in `ros_ws/src/hw01_tf_demo/`, broadcast the body frame and space frame as TF frames for the same motion as Task 1, and animate it in `rviz2`. Add a toggle (a ROS parameter or a keyboard input) that switches between composing the next rotation about the current frame vs. the fixed frame, so the difference is visible live rather than as two separate recordings.

AI Use Note (required if AI was used on Part 2)

Tool used:

What I asked it:

How I used the output (and what I changed/verified myself):

Problem 1. Vector and cross-product identities

Source: Murray, Li & Sastry (1994), Ch. 2, Exercise 1

Let $a, b, c \in \mathbb{R}^3$ be 3-vectors and let $\cdot$ and $\times$ denote the dot product and cross product in $\mathbb{R}^3$. Verify the following identities:

$$(a) \quad a \cdot (b \times c) = (a \times b) \cdot c$$

$$(b) \quad a \times (b \times c) = (a \cdot c)b - (a \cdot b)c$$

$$\Rightarrow a = [a_1 \ a_2 \ a_3]^T$$

$$b = [b_1 \ b_2 \ b_3]^T$$

$$c = [c_1 \ c_2 \ c_3]^T.$$

$$b \times c = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{bmatrix} b_2 c_3 - b_3 c_2 \\ b_3 c_1 - b_1 c_3 \\ b_1 c_2 - b_2 c_1 \end{bmatrix}.$$

$$\vec{a} \cdot \vec{b} \times \vec{c} = a_1(b_2 c_3 - b_3 c_2) + a_2(b_3 c_1 - b_1 c_3) + a_3(b_1 c_2 - b_2 c_1)$$

$$= c_1(a_2 b_3 - a_3 b_2) + c_2(a_3 b_1 - a_1 b_3) + c_3(a_1 b_2 - a_2 b_1).$$

$$= \vec{c} \cdot (\vec{a} \times \vec{b}).$$

Scalar triple product is interchangeable.

Tool used $\rightarrow$ Mind, only. 😊

$$b) \vec{a} \times (\vec{b} \times \vec{c}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_2c_3 - b_3c_2 & b_3c_1 - b_1c_3 & b_1c_2 - b_2c_1 \end{vmatrix}$$

$$\hat{i}[(a_2c_3b_1 - a_2b_2c_1) - (a_3b_3c_1 - a_3c_3b_1)] -$$

$$\hat{j}[(a_1b_1c_2 - a_1c_1b_2) - (a_3c_3b_2 - a_3b_3c_2)] +$$

$$\hat{k}[(a_1c_1b_3 - a_1b_1c_3) - (a_2b_2c_3 - a_2c_2b_3)].$$

Problem 2. Frames, rotation matrices, and what they represent

Source: Lynch & Park (2017), Ch. 3, Exercise 3.1(a-h)

In terms of the $\hat{x}_s, \hat{y}_s, \hat{z}_s$ coordinates of a fixed space frame $\{s\}$, the frame $\{a\}$ has its $\hat{x}_a$ -axis pointing in the direction $(0, 0, 1)$ and its $\hat{y}_a$ -axis pointing in the direction $(-1, 0, 0)$, and the frame $\{b\}$ has its $\hat{x}_b$ -axis pointing in the direction $(1, 0, 0)$ and its $\hat{y}_b$ -axis pointing in the direction $(0, 0, -1)$.

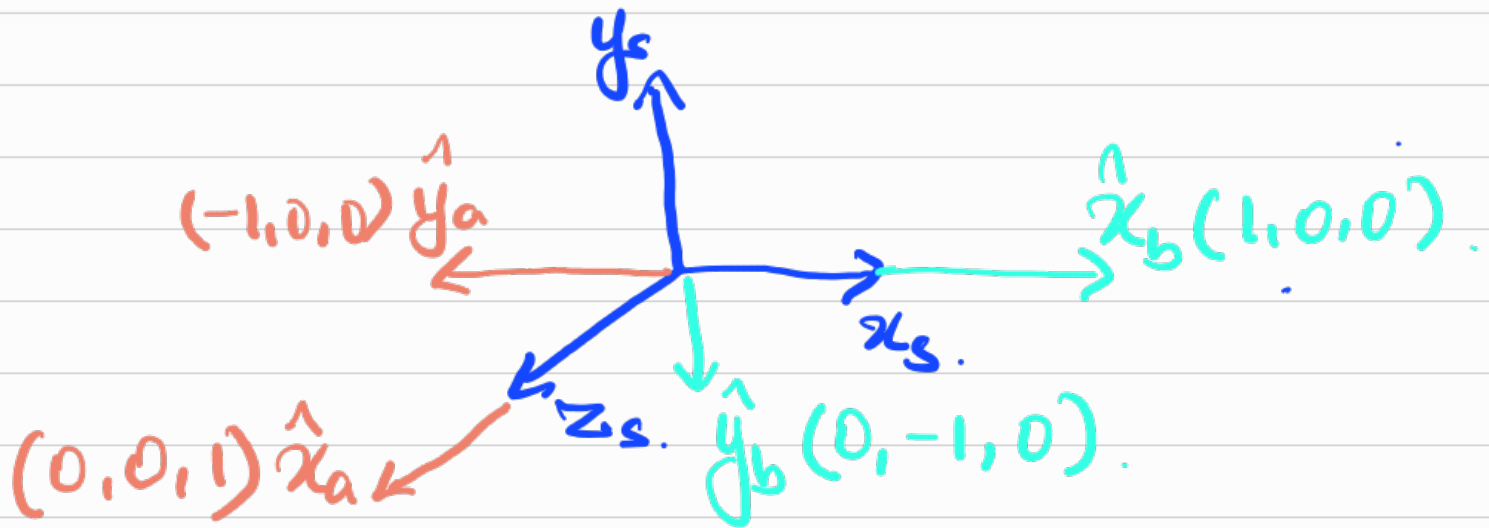
Generated with AI; Curated using HI.

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(a)



(b) Write down the rotation matrices R_{sa} and R_{sb} .

(b) Take CCW as +ve.

turn by 90° , CCW $\rightarrow \phi = +90$
turn by 90° ; CW.
 $\hookrightarrow \theta = -90$.

$$R_{sa} = R_{sz} \circ R_{sy}$$

$$= \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$R_{sa} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

Similarly,

$$R_{sb} = R_{sx} \rightarrow \theta = 180^\circ, \text{ CCW.}$$

$$R_{sb} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$R_{sb} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

(b) Write down the rotation matrices R_{sa} and R_{sb} .

(c) Given R_{sb} , how do you calculate R_{sb}^{-1} without using a matrix inverse? Write down R_{sb}^{-1} and verify its correctness using your drawing.

$$(c) \because R_{sb} \rightarrow SO(3).$$

$$R_{sb}^{-1} = R_{sb}^T.$$

$$\Rightarrow R_{sb}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}.$$

$$\hookrightarrow \because \text{Symmetric } R_{sb} = R_{sb}^T$$

(d) Given R_{sa} and R_{sb} , how do you calculate R_{ab} (again without using matrix inverses)? Compute the answer and verify it using your drawing.

$$(d) R_{sb} = R_{ab} R_{sa}$$

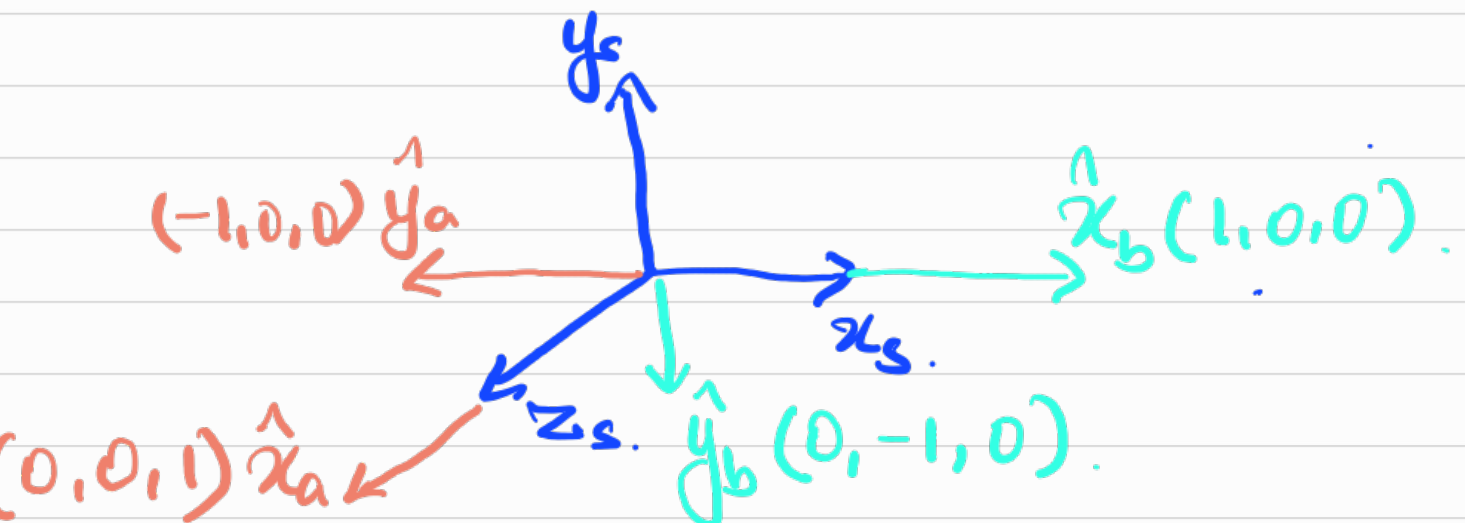
Post multiply R_{sa}^T

$$R_{sb} R_{sa}^T = R_{ab} \times I.$$

$$\Rightarrow R_{ab} = R_{sb} R_{sa}^T$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$$

$$R_{ab} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$



For instance take $\hat{x}_a = [0 \ 0 \ 1]^T$

Take $\rightarrow R_{ab} \hat{x}_a$.

$$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} +1 \\ 0 \\ 0 \end{bmatrix} = \hat{x}_b$$

Take $R_{ab} \hat{y}_a$

$$= \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} = \hat{y}_b$$

Hence Proved.

the answer and verify it using your drawing.

- (e) Let $R = R_{sb}$ be considered as a transformation operator consisting of a rotation about $\hat{x}$ by -90° . Calculate $R_1 = R_{sa}R$, and think of R_{sa} as a representation of an orientation, R as a rotation applied to R_{sa} , and R_1 as the new orientation after the rotation has been performed. Does the new orientation R_1 correspond to a rotation of R_{sa} by -90° about the world-fixed $\hat{x}_s$ -axis or about the body-fixed $\hat{x}_a$ -axis? Now calculate $R_2 = RR_{sa}$. Does R_2 correspond to a rotation of R_{sa} by -90° about the world-fixed $\hat{x}_s$ -axis or about the body-fixed $\hat{x}_a$ -axis?

$\Rightarrow R = \text{Rotation about } \hat{x} \text{ by } -90^\circ.$

Computed R_{sa} .

$$R_{sa} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(-90^\circ) & -\sin(-90^\circ) \\ 0 & \sin(-90^\circ) & \cos(-90^\circ) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

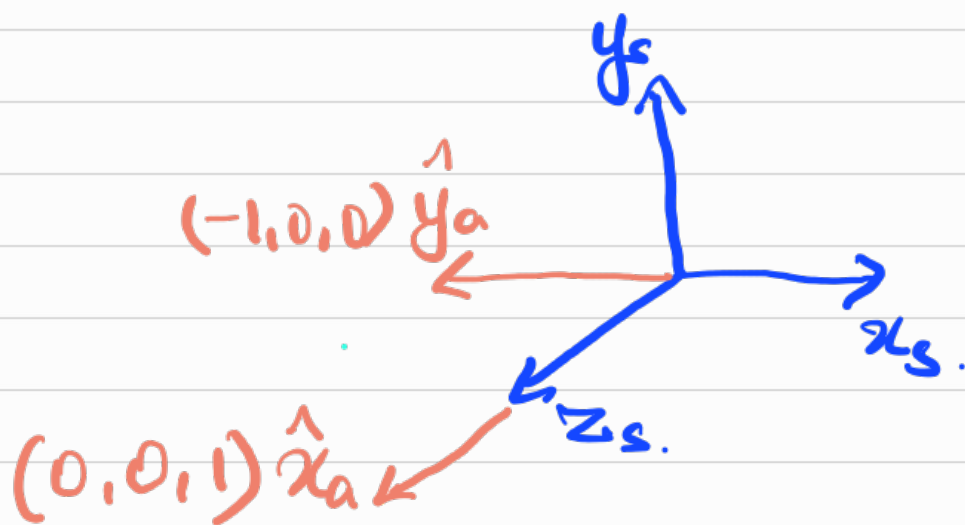
Given,

$$R_1 = R_{sa}R$$

$$= \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

CCW +ve.



$$\hat{x}_{a_{new}, s} = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\hat{y}_{a_{new}, s} = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$\Rightarrow$ Rotation about x_a
post multiply

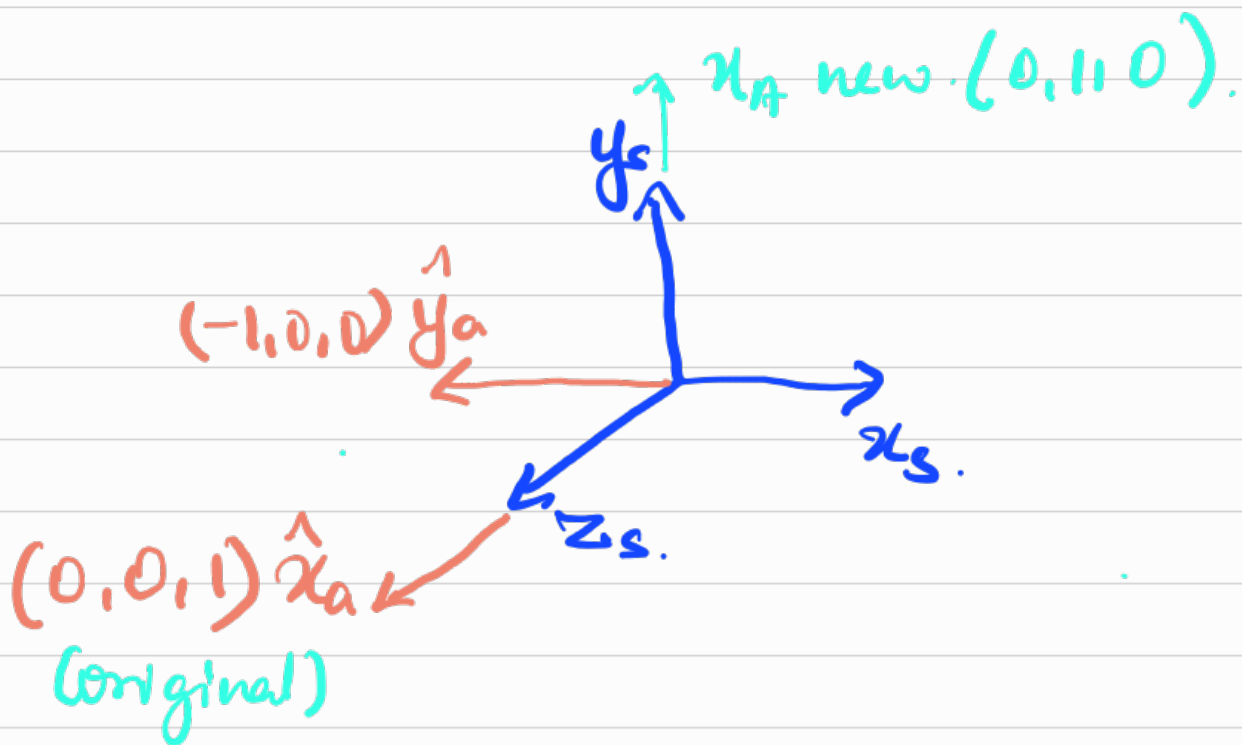
$x_s \rightarrow x_{a \text{ original}} \rightarrow \text{no change}$

$y_s \rightarrow y_{a \text{ original}} \rightarrow \text{yes change}$

Now $R_2 = R R_{sa}$.

$$R_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$R_2 = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



$$x_{2\text{new}} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$
$$= \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$x_s \longrightarrow x_\theta \longrightarrow x_{Anew2}$$

CW rotation about x_s axis.

So ' R_2 ' represents the rotation about global x_s .

$$y_{dnew} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow y_{\text{original}}$$

So no change when rotated about global x -axis.

Hence Proved.

- (f) Use R_{sb} to change the representation of the point $p_b = (1, 2, 3)$ (given in $\{b\}$ coordinates) to $\{s\}$ coordinates.

$$p_s = R_{sb} p_b = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$= \underline{\begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}}$$

(g) coordinates:

- (g) Choose a point p represented by $p_s = (1, 2, 3)$ in $\{s\}$ coordinates. Calculate $p' = R_{sb} p_s$ and $p'' = R_{sb}^T p_s$. For each operation, should the result be interpreted as *changing coordinates* (from the $\{s\}$ frame to $\{b\}$) without moving the point p , or as *moving the location* of the point without changing the reference frame of the representation?

$$\Rightarrow p' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$$

$$p'' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$$

$p' \rightarrow$ Moving location of point without changing the frame of reference.

$p'' \rightarrow$ Moving reference frame from $\{s\}$ to $\{b\}$

without moving the vector.

Source of point of answer → Not able to comprehend
So answer is from AI

Physical Interpretation

- Operation $p' = R_{sb}p_s$:
 - Interpretation: Moving the location of the point without changing the reference frame of the representation.
 - Reasoning: In rigid-body kinematics, multiplying a coordinate vector by a rotation matrix within the same frame acts as a **rotation operator**. The physical point p is rotated to a new position in space, and the resulting coordinates p' are still represented with respect to the original $\{s\}$ frame.
- Operation $p'' = R_{sb}^T p_s$:
 - Interpretation: Changing coordinates (from the $\{s\}$ frame to $\{b\}$) without moving the point p .
 - Reasoning: The rotation matrix R_{sb} transforms coordinates from frame $\{b\}$ to frame $\{s\}$ according to $p_s = R_{sb}p_b$. Because rotation matrices are proper orthogonal matrices ($R_{sb}^{-1} = R_{sb}^T$), solving for p_b yields $p_b = R_{sb}^T p_s$. Therefore, $p'' = p_b$ represents the coordinates of the *unmoved* point p viewed relative to frame $\{b\}$.

(h) An angular velocity ω is represented in $\{s\}$ as $\omega_s = (3, 2, 1)$. What is its representation ω_a in $\{a\}$?

$$\Rightarrow \omega_s = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

$$\omega_a = R_{sa} \omega_s.$$

$$= \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ -3 \\ 1 \end{bmatrix}$$

Let p be a point whose coordinates are $p = \left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{2}} \right)$ with respect to the fixed frame $\hat{x}-\hat{y}-\hat{z}$. Suppose that p is rotated about the fixed-frame $\hat{x}$ -axis by 30° , then about the fixed-frame $\hat{y}$ -axis by 135° , and finally about the fixed-frame $\hat{z}$ -axis by -120° . Denote the coordinates of this newly rotated point by p' .

(a) What are the coordinates p' ?

$$\Rightarrow \begin{array}{l} 1^{\text{st}} \rightarrow R_x \rightarrow 30^\circ \\ 2^{\text{nd}} \rightarrow R_y \rightarrow 135^\circ \\ 3^{\text{rd}} \rightarrow R_z \rightarrow -120^\circ \end{array}$$

$$R_{\text{comb}} = R_z R_y R_x$$

$$= \begin{bmatrix} \frac{1}{2} & +\frac{\sqrt{3}}{2} & 0 \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2}\sqrt{2} & \frac{6-\sqrt{2}}{8} & \frac{-2\sqrt{3}-\sqrt{6}}{8} \\ -\frac{\sqrt{6}}{4} & \frac{2\sqrt{3}+\sqrt{6}}{8} & \frac{3\sqrt{2}-2}{8} \\ \frac{1}{\sqrt{2}} & \frac{1}{2}\sqrt{2} & \frac{\sqrt{6}}{4} \end{bmatrix}$$

$$p' = R_{\text{comb}} p = R_{\text{comb}} \begin{bmatrix} \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$p' = \begin{bmatrix} -0.5938 \\ -0.5101 \\ 0.6711 \end{bmatrix}$$

(b) Find the rotation matrix R such that $p' = Rp$ for the p' you obtained in (a).

Already Found

$$R_{\text{comb}} = \begin{bmatrix} \frac{1}{2}\sqrt{2} & \frac{6-\sqrt{2}}{8} & \frac{-2\sqrt{3}-\sqrt{6}}{8} \\ -\sqrt{6}/4 & \frac{2\sqrt{3}+\sqrt{6}}{8} & \frac{3\sqrt{2}-2}{8} \\ \frac{1}{4}\sqrt{2} & \frac{1}{2}\sqrt{2} & \sqrt{6}/4 \end{bmatrix}.$$

Problem 4. Proving $R_a^b R_b^c = R_a^c$

Source: Lynch & Park (2017), Ch. 3, Exercise 3.4

In this exercise you are asked to prove the property $R_a^b R_b^c = R_a^c$ used in class. Define the unit axes of frames $\{a\}$, $\{b\}$, and $\{c\}$ by the triplets of orthogonal unit vectors $\{\hat{x}_a, \hat{y}_a, \hat{z}_a\}$, $\{\hat{x}_b, \hat{y}_b, \hat{z}_b\}$, and $\{\hat{x}_c, \hat{y}_c, \hat{z}_c\}$, respectively. Suppose that the unit axes of frame $\{b\}$ can be expressed in terms of the unit axes of frame $\{a\}$ by,

$$\hat{x}_b = r_{11}\hat{x}_a + r_{21}\hat{y}_a + r_{31}\hat{z}_a,$$

$$\hat{y}_b = r_{12}\hat{x}_a + r_{22}\hat{y}_a + r_{32}\hat{z}_a,$$

$$\hat{z}_b = r_{13}\hat{x}_a + r_{23}\hat{y}_a + r_{33}\hat{z}_a.$$

Generated with AI; Curated using HI.

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Similarly, suppose that the unit axes of frame $\{c\}$ can be expressed in terms of the unit axes of frame $\{b\}$ by

$$\hat{x}_c = s_{11}\hat{x}_b + s_{21}\hat{y}_b + s_{31}\hat{z}_b,$$

$$\hat{y}_c = s_{12}\hat{x}_b + s_{22}\hat{y}_b + s_{32}\hat{z}_b,$$

$$\hat{z}_c = s_{13}\hat{x}_b + s_{23}\hat{y}_b + s_{33}\hat{z}_b.$$

From the above, prove that $R_a^b R_b^c = R_a^c$.

Problem 5. Properties of rotation matrices and axes

Given,

$$\begin{bmatrix} \hat{x}_b \\ \hat{y}_b \\ \hat{z}_b \end{bmatrix} = R_a^b \begin{bmatrix} \hat{x}_a \\ \hat{y}_a \\ \hat{z}_a \end{bmatrix}$$

$$\begin{bmatrix} x_b \\ y_b \\ z_b \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} x_a \\ y_a \\ z_a \end{bmatrix} \quad \text{--- (i)}$$

$$\Rightarrow \hat{x}_b = r_{11} \hat{x}_a + r_{12} \hat{y}_a + r_{13} \hat{z}_a$$

$$\Rightarrow \hat{y}_b = r_{21} \hat{x}_a + r_{22} \hat{y}_a + r_{23} \hat{z}_a$$

$$\Rightarrow \hat{z}_b = r_{31} \hat{x}_a + r_{32} \hat{y}_a + r_{33} \hat{z}_a.$$

Similarly

$$\begin{bmatrix} \hat{x}_c \\ \hat{y}_c \\ \hat{z}_c \end{bmatrix} = R_b^c \begin{bmatrix} \hat{x}_b \\ \hat{y}_b \\ \hat{z}_b \end{bmatrix}$$

$$R_b^c = \begin{bmatrix} s_{11} & s_{12} & s_{13} \\ s_{21} & s_{22} & s_{23} \\ s_{31} & s_{32} & s_{33} \end{bmatrix}$$

$$\begin{bmatrix} \hat{x}_c \\ \hat{y}_c \\ \hat{z}_c \end{bmatrix} = \begin{bmatrix} s_{11} & s_{12} & s_{13} \\ s_{21} & s_{22} & s_{23} \\ s_{31} & s_{32} & s_{33} \end{bmatrix} \begin{bmatrix} x_b \\ y_b \\ z_b \end{bmatrix} \quad \text{--- (ii)}$$

So,

$$\hat{x}_c = S_{11}\hat{x}_b + S_{12}\hat{y}_b + S_{13}\hat{z}_b$$

$$\hat{y}_c = S_{21}\hat{x}_b + S_{22}\hat{y}_b + S_{23}\hat{z}_b$$

$$\hat{z}_c = S_{31}\hat{x}_b + S_{32}\hat{y}_b + S_{33}\hat{z}_b$$

Substitute eq(i) in eq(ii)

$$\begin{bmatrix} \hat{x}_c \\ \hat{y}_c \\ \hat{z}_c \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} \hat{x}_a \\ \hat{y}_a \\ \hat{z}_a \end{bmatrix}$$

$$= \begin{bmatrix} S_{11}r_{11} + S_{12}r_{21} + S_{13}r_{31} & S_{11}r_{12} + S_{12}r_{22} + S_{13}r_{32} & \dots \\ S_{21}r_{11} + S_{22}r_{21} + S_{23}r_{31} & \dots & \dots \\ S_{31}r_{11} + S_{32}r_{21} + S_{33}r_{31} & \dots & \dots \end{bmatrix}$$

$$\downarrow R_c^a$$

$$\begin{bmatrix} \hat{x}_a \\ \hat{y}_a \\ \hat{z}_a \end{bmatrix}$$

Thus,

$$\boxed{R_c^a = R_b^c R_a^b}$$

Problem 5. Properties of rotation matrices and $\mathfrak{so}(3)$

Source: Murray, Li & Sastry (1994), Ch. 2, Ex. 3(c) & Ex. 4(a-c)

Let $R \in SO(3)$.

- (a) [Recall from lecture] Show that for $v, w \in \mathbb{R}^3$,

$$R(v \times w) = (Rv) \times (Rw).$$

This was derived in class – write the argument again in your own words (no need to search for a new proof).

- (b) For $\omega \in \mathbb{R}^3$, show that

$$R \hat{\omega} R^T = \widehat{(R\omega)}.$$

- (c) Let $R = [r_1 \ r_2 \ r_3]$ be a rotation matrix written in terms of its columns. Show that

$$\det R = r_1^T(r_2 \times r_3).$$

- (d) Show that $\mathfrak{so}(3)$ (the set of 3×3 skew-symmetric matrices) is a vector space. Determine its dimension and give a basis.

$$\Rightarrow \bar{v} = [v_1 \ v_2 \ v_3]$$

$$\bar{w} = [w_1 \ w_2 \ w_3]$$

$$\bar{v} \times \bar{w} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = \begin{bmatrix} v_2 w_3 - v_3 w_2 \\ -v_1 w_3 + v_3 w_1 \\ v_1 w_2 + v_2 w_1 \end{bmatrix}_{3 \times 1}$$

W.K.T all rotation can be split into 3 basic rotation about x-, y-, z-

Thus,
for simplification purpose I will use rotation R_2 for now.

Problem changes to
 $R_2(\bar{v} \times \bar{w}) = R_2 \bar{v} \times R_2 \bar{w}.$

$$R_2(\bar{v} \times \bar{w}) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} v_2 \omega_3 - v_3 \omega_2 \\ -v_1 \omega_3 + v_3 \omega_1 \\ v_1 \omega_2 + v_2 \omega_1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta [v_2 \omega_3 - v_3 \omega_2] + \sin\theta (-v_1 \omega_3 + v_3 \omega_1) \\ \sin\theta (v_2 \omega_3 - v_3 \omega_2) - \cos\theta (-v_1 \omega_3 + v_3 \omega_1) \\ v_1 \omega_2 + v_2 \omega_1 \end{bmatrix}$$

$$R_2 \bar{v} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$$R_2 \bar{v} = \begin{bmatrix} \cos\theta v_1 - \sin\theta v_2 \\ \sin\theta v_1 + \cos\theta v_2 \\ v_3 \end{bmatrix}$$

$$R_2 \bar{w} = \begin{bmatrix} \cos\theta \omega_1 - \sin\theta \omega_2 \\ \sin\theta \omega_1 + \cos\theta \omega_2 \\ \omega_3 \end{bmatrix}$$

$$R_2 \bar{v} \times R_2 \bar{w}$$

$$\begin{array}{c} \begin{array}{ccc} i & j & k \\ \left| \begin{array}{ccc} c\theta v_1 - s\theta v_2 & s\theta v_1 + c\theta v_2 & v_3 \\ c\theta w_1 - s\theta w_2 & s\theta w_1 + c\theta w_2 & w_3 \end{array} \right| \end{array} \end{array}$$

$$\begin{array}{c} \left\{ \begin{array}{l} s\theta v_1 w_3 + c\theta v_2 w_3 - s\theta v_3 w_1 + c\theta v_3 w_2 \\ - (c\theta v_1 w_3 - s\theta v_2 w_3 + c\theta w_1 v_3 - s\theta w_2 v_3) \\ c\theta c\theta w_1 + c^2\theta v_1 w_2 - s^2\theta v_2 w_1 - s\theta c\theta v_2 w_2 \\ - (s\theta c\theta v_1 w_1 + c^2\theta v_2 w_1 - s^2\theta v_1 w_2 - c\theta s\theta v_2 w_2) \end{array} \right\} \end{array}$$

$$= \begin{array}{c} \left| \begin{array}{l} c\theta(v_3 w_2 + v_2 w_3) + s\theta(v_1 w_3 - v_3 w_1) \\ s\theta(v_2 w_3 - v_3 w_2) - c\theta(v_1 w_3 - v_3 w_1) \\ v_1 w_2 + v_2 w_1 \end{array} \right| \end{array}$$

$$= R(\bar{v} \times \bar{w}).$$

$$(b). \vec{\omega} = [\omega_1 \ \omega_2 \ \omega_3]$$

$$\hat{\omega} = \begin{vmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{vmatrix}$$

Let's take arbitrary vector $\vec{a}$

$$R \hat{\omega} R^T \vec{a} = R \hat{\omega} \vec{x}$$

$$= R(\vec{\omega} \times \vec{x})$$

$$= R\vec{\omega} \times R\vec{x}$$

$$= R\vec{\omega} \times R R^T \vec{a}$$

$$= \underbrace{R\vec{\omega}}_{\text{also vector}} \times \vec{a}$$

$$R \hat{\omega} R^T \vec{a} = R \hat{\omega} \vec{a}$$

Comparing LHS and RHS.

$$R \hat{\omega} R = R \hat{\omega}$$

Tool used : Discussion with IIT buddy.

(c) Rotation matrix.

$$R = [\bar{r}_1 \quad \bar{r}_2 \quad \bar{r}_3]$$

$$\bar{r}_1 = [r_{11} \quad r_{12} \quad r_{13}]^T$$

$$\bar{r}_2 = [r_{21} \quad r_{22} \quad r_{23}]^T$$

$$\bar{r}_3 = [r_{31} \quad r_{32} \quad r_{33}]^T$$

$$\det R = \begin{vmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{vmatrix} = r_{11}(r_{22}r_{33} - r_{23}r_{32}) - r_{21}(r_{12}r_{33} - r_{32}r_{13}) + r_{31}(r_{12}r_{23} - r_{22}r_{13}).$$

$$= [r_{11} \quad r_{21} \quad r_{31}] \begin{matrix} 1 \times 3 \\ 3 \times 1 \end{matrix} \begin{bmatrix} r_{22}r_{33} - r_{23}r_{32} \\ r_{12}r_{33} - r_{32}r_{13} \\ r_{12}r_{23} - r_{22}r_{13} \end{bmatrix}$$

$\hookrightarrow r_2 \times r_3.$

$$= \underline{\underline{r_1^T (r_2 \times r_3)}}.$$

(d) Given.
 $SO(3)$ skew symmetric $\Rightarrow A^T = -A = S$.

Consider $S = \begin{bmatrix} 0 & -a_{21} & -a_{31} \\ a_{21} & 0 & -a_{32} \\ a_{31} & a_{32} & 0 \end{bmatrix}$

Then $-S = S^T$

$\Rightarrow S = \begin{bmatrix} 0 & a_{21} & a_{31} \\ -a_{21} & 0 & a_{32} \\ -a_{31} & -a_{32} & 0 \end{bmatrix}$.

Stuck. Tool Used - Gemini Pro.

What it is telling: to prove any matrix is a vector space.
I need to be able to break the matrix into a set of linear combination matrices

these operations that we turn V into a vector space.

Definition 4.1.1. A **vector space** over $\mathbb{F}$ is a set V together with the operations of addition $V \times V \rightarrow V$ and scalar multiplication $\mathbb{F} \times V \rightarrow V$ satisfying each of the following properties.

1. **Commutativity:** $u + v = v + u$ for all $u, v \in V$;
2. **Associativity:** $(u + v) + w = u + (v + w)$ and $(ab)v = a(bv)$ for all $u, v, w \in V$ and $a, b \in \mathbb{F}$;
3. **Additive identity:** There exists an element $0 \in V$ such that $0 + v = v$ for all $v \in V$;
4. **Additive inverse:** For every $v \in V$, there exists an element $w \in V$ such that $v + w = 0$;
5. **Multiplicative identity:** $1v = v$ for all $v \in V$;
6. **Distributivity:** $a(u + v) = au + av$ and $(a + b)u = au + bu$ for all $u, v \in V$ and $a, b \in \mathbb{F}$.

A **vector space** over $\mathbb{R}$ is usually called a **real vector space**, and a vector space over $\mathbb{C}$ is similarly called a

Attached above is the source it is referring to.

Here is the AI response.

Why should i prove if i am able to decompose linearly then it is a vector space?

- Expressing a set as a linear combination of fixed matrices proves that the set is a "span" [[Span Definition \(LibreTexts\)](#)].
- Mathematical theorems state that the span of any subset within a known vector space (such as 3×3 matrices) inherently forms a valid subspace, automatically satisfying vector space axioms like closure [[Subspaces Theorem \(LibreTexts\)](#)].
- Therefore, proving the linear decomposition theoretically establishes the set as a vector space without manually testing closure rules [[Subspaces Theorem \(LibreTexts\)](#)].
- However, standard academic grading rubrics expect students to explicitly demonstrate closure under addition and scalar multiplication when a question isolates "Show it is a vector space" from "Give a basis" [[Vector Space Axioms \(Wolfram MathWorld\)](#)].

Thus,
I break 'S' into,

$$S = a_{21} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + a_{31} \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\Downarrow$$

$$S_{21} + a_{32} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \quad \Downarrow$$

$$\Downarrow S_{32} \quad S_{31}$$

Note that S_{21} , S_{31} , S_{32} are linearly independent.

Thus S_{21} , S_{31} and S_{32} are the basis vectors for 'S'.

Problem 6. The planar rotation group $SO(2)$

Source: Murray, Li & Sastry (1994), Ch. 2, Exercise 10(a,c)

Let $SO(2)$ be the set of all 2×2 orthogonal matrices with determinant equal to $+1$.

(a) Show that $SO(2)$ can be identified with $\mathbb{S}^1$, the unit circle in $\mathbb{R}^2$.

(b) Show that for $R \in SO(2)$ and $\hat{\omega} \in \mathfrak{so}(2)$,

$$R \hat{\omega} R^T = \hat{\omega}.$$

Only fill in the note below if you used AI on a non-flagged part of Part 1 (most students will leave this one blank).

(a) If $R \in SO(2)$.

$$R = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

W.K.T

$$ad - bc = 1$$

Also

$$R^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

$$RR^T = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

$$a^2 + b^2 = 1$$

$$c^2 + d^2 = 1$$

$$ad - bc = 1.$$

$$\Rightarrow R = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$$

$$\cos^2 \theta + \sin^2 \theta = 1 \Rightarrow a^2 + b^2.$$

$$\cos^2 \theta + (-\sin \theta)^2 = 1 \Rightarrow c^2 + d^2.$$

$$\cos^2 \theta + \sin^2 \theta = 1 \Rightarrow c^2 + d^2.$$

equation of unit circle.

$$(b) R \hat{\omega} R^T,$$

for 2x rotation matrices

$$\hat{\omega} = \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix}.$$

Thus

$$R \hat{\omega} = \hat{\omega} R.$$

$$\Rightarrow R \hat{\omega} R^T = \hat{\omega} \underbrace{R R^T}_{I} \\ = \hat{\omega}.$$

Hence Proved.